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REVIEW ARTICLE

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Model-based systems engineering: Evaluating perceived value, metrics, and evidence through literature

Kelly X. Campo | Thomas Teper | Casey E. Eaton | Anna M. Shipman | Garima Bhatia | Bryan Mesmer

Industrial & Systems Engineering and Engineering Management Department, The University of Alabama in Huntsville, Huntsville, Alabama, USA

Correspondence
Kelly X. Campo, Industrial & Systems Engineering and Engineering Management Department, The University of Alabama in Huntsville, 301 Sparkman Dr, Huntsville, AL, USA.
Email: kxc0001@uah.edu

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Abstract

Although Model-Based Systems Engineering (MBSE) is quickly becoming adopted in Systems Engineering (SE), there have not been many studies evaluating literature to determine the perceived value of implementing MBSE. This research first identifies and discusses previous studies on the justification or rejection of MBSE. This study investigates how the SE community perceives the value of MBSE by coding and analyzing positive and negative descriptions of MBSE; perceived benefits and drawbacks of implementing MBSE; and the evidence and metrics used to substantiate and measure each claim about MBSE. From 60 academic sources, this study collected and coded over 2900 claims on MBSE. Our findings determine the most positive attributes of MBSE to be *Verification & Validation Capability*, *Consistency*, *Reasoning*, and *Risk & Error Manageability*, while the most negative attributes were *Approach Understandability*, *Acceptability*, *Familiarity*, and *Approach Complexity*. The most-stated benefits were *Reduced Time*, *Better Communication/Information Sharing*, *Reduced Costs*, and *Better Analysis Capability*. The most claimed drawbacks were *Increased Costs*, *Increased Time*, *Increased Effort*, and *Worsened Capability*. A large share of claims (47%) about MBSE was based on author opinions. Most claims (86%) were not substantiated by a metric.

KEYWORDS
MBSE, model-based systems engineering, systems engineering, value

1 | INTRODUCTION

1.1 | Motivation and research questions

Differentiating a fad from a paradigm shift can be a significant challenge for systems engineers. The choice of whether to try a new approach on an engineering project can result in improved or worsened budgets, schedules, efficiency, safety, and morale.[1] In the midst of an industry-wide Digital Engineering wave, systems engineers are increasingly facing this question with regards to adopting Model-Based Systems Engineering (MBSE). To make this decision even more challenging, the engineering projects that are supposed to gain the most from MBSE are high stakes and difficult to experiment on.[2] This study gathers over 2900 claims about MBSE with perceived positive and negative aspects from 60 academic sources. The perceived positive and negative claims about MBSE are aggregated along with the type of evidence used to substantiate the claim, the metrics used to measure the claims, and the perceived benefits and drawbacks of MBSE to provide a formalized understanding of what is perceived as valuable about the approach and how that value can be measured.

A recent article examining the adoption of MBSE and its driving or hindering factors found that practitioners could not identify which

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aspects of MBSE contributed to the expected benefits, concluding that poor experiences and frustrations with MBSE are a result of false or lofty expectations.[3] This indicates that the aggregation methodology of the study in this article identifying positive and negative aspects of MBSE offers significant progress towards understanding the value of MBSE. We aim to improve, over past studies, the insights as to the value of MBSE through addressing the following research questions:

RQ1: What attributes and impacts of MBSE are perceived positively or negatively in literature?

RQ2: How are these attributes and impacts substantiated through evidence in literature?

RQ3: How are these attributes and impacts of MBSE measured in literature?

In this paper, the background section describes MBSE and provides a summary of previous similar studies on the value of MBSE, contextualizing the current state of research on the value of MBSE. The methodology section details the steps taken in this study to gather data (quotes from literature) in a systematic manner. The coding method used to standardize the gathered quotes (called claims hereafter) into analyzable data is also described. The results and discussion section presents the findings, identifying what aspects of MBSE are perceived positively or negatively by the Systems Engineering (SE) community and how those aspects are measured with metrics and substantiated with types of evidence. The conclusion section contextualizes the findings, identifies the limitations of this study, and provides recommendations for future research.

1.2 | Background

In recent years, the Department of Defense has prioritized developing new capabilities at record speeds, largely through a focus on managing evolutionary complex systems through Digital Engineering (DE).[4] This focus on DE has expanded to other organizations like NASA, which has prioritized the transition to DE operations in order to "achieve the highest level of collaboration, interoperability, efficiency, and data availability" in projects.[5] NASA defines a DE environment as "an environment that encompasses the collaboration of digital project and technical product data among participants" by enabling interaction through digital tools and technologies built on model-based enterprise methods and practices.[5] In general, DE is defined as "an integrated digital approach that uses authoritative sources of systems data and models as a continuum across disciplines to support lifecycle activities from concept through disposal.[4]" With the emphasis on DE, model-based methodologies, like MBSE, are increasingly considered for various industry implementations.[4]

The International Council on Systems Engineering (INCOSE) defines MBSE as "the formalized application of modeling to support system requirements, design, analysis, verification and validation activities. MBSE begins in the conceptual design phase and continues throughout development and later life cycle phases.[6]" MBSE is a SE approach that includes "behavioral analysis, system architecture, requirement traceability, performance analysis, simulation, test, etc.[3]" MBSE can be used as an alternative to Document-Based Systems Engineering (DBSE). While DBSE uses documents as the primary source of information exchange, MBSE focuses on domain models as the primary source of information exchange. MBSE combines models—simplifications of a system or component—with systems thinking—the process of developing a system with special attention to relationships between different subsystems that form a "coherent whole[7]".

When the MBSE approach is leveraged in industry, the Object Management Group's (OMG) Systems Modeling Language (SysML) is often used.[8] SysML was developed as a subset of Unified Modeling Language 2 (UML 2) jointly by the OMG and INCOSE in 2003 and is commonly used as the graphical modeling language of choice for MBSE applications.[9] Other languages for MBSE applications include Eclipse's Arcadia, Electronics Architecture and Software Technology Architecture Description Language (EAST-ADL), Knowledge Based Systems, Inc.'s Integration DEFinition (IDEF), and ISO's Object Process Methodology (OPM).[10]

Major sectors like aerospace, defense, and complex system product engineering have implemented MBSE.[2,11,12] Other industries like transportation, energy, financial, architecture, consumer retail, medical, etc. have also begun exploring and implementing MBSE.[1,2,12] MBSE is often lauded as the "future of SE" due to its ability to model an

SIGNIFICANCE AND PRACTITIONER POINTS

This research is novel in that no previous studies have tried to evaluate the value of Model-Based Systems Engineering (MBSE) by analyzing attributes, benefits, and drawbacks, while also evaluating the types of evidence and metrics used to support the claims in literature. Practitioners need justification in determining if MBSE is suitable for their applications which requires an understanding of MBSE's strengths and weaknesses. This study identifies 68 attributes describing MBSE, enabling practitioners to evaluate these attributes in regard to their applications and to understand which benefits are more supported with metrics and evidence. This work identifies high positive perceptions of MBSE in the systems engineering community and significantly more perceived benefits than perceived drawbacks. However, a large portion of claims regarding MBSE is not yet supported with justifiable evidence. Attributes with both negative and positive perceptions are identified in this research, providing researchers with specific claims on MBSE that require more investigation to justify their promotion. Practitioners should be cautious of relying solely on reported benefits to base their decisions about MBSE due to the significant lack of quantitative metrics and justifiable evidence currently available.

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TABLE 1 Fifteen unique terms used to identify literature sources

Search terms used		
MBSE	MBSE versus DBSE	Value of MBSE literature
MBSE adoption	Model-based systems	MBSE in literature
MBSE case study	Model-based systems engineering	MBSE quantifying value
Critical analysis of MBSE	Perceptions of MBSE	Value of MBSE
MBSE aerospace	MBSE systematic literature review	Model-based systems engineering NASA

entire complex system, and many in the SE community believe it will soon become the status quo.[13,14] Despite a significant wave of positive perceptions about the approach, it is still unclear what is contributing to the popularity of MBSE. Is there an identifiable value to MBSE (and what is the value), or is MBSE simply "much ado about nothing new"?[13]

1.2.1 | Previous studies seeking to understand the value of MBSE

While MBSE is a popular topic in SE, few studies have considered how MBSE is justified or how to understand the value of MBSE from a community viewpoint. The similar studies are discussed below.

Previous studies' methodologies

Our studyaggregates attributes and metrics about MBSE from a variety of data sources, providing a comprehensive, evidence-based public literature analysis on the topic. This methodology is different from the many identified studies on the value of MBSE, which use interviews and surveys of professionals experienced with MBSE.[3,12,15–17] The researchers use survey or interview data to support the implementation of MBSE or to identify challenges prior to the implementation of MBSE. Specific attributes used to justify MBSE include reduced complexity and increased system lifespan, profitability, applicability, consistency, and efficiency.[11,12,17–19]

Understanding the negative and positive about MBSE

Many studies focus on identifying only positive attributes of MBSE. This may bias the understanding of the value of MBSE. As a result, many of the previous studies on the value of MBSE advocate for its adoption.[3,11,15,16,18–21] Only one identified similar study directly rejected the adoption of MBSE in favor of a document-based approach.[13] Few of the similar studies attempt to identify both the benefits and drawbacks of MBSE.[3,11,15,19] The study presented in this article analyzes both the benefits and drawbacks of MBSE to obtain a holistic understanding.

Understanding the value of MBSE economically

A few studies attempt to economically justify the adoption of MBSE by analyzing the costs and benefits of MBSE.[2,16,19,20] Dabkowski et al.[20] apply a constructive SE cost model that estimates the cost of adding an additional subsystem to an existing system in order to justify the costs of MBSE. Madni and Purohit[2] create a cost comparison chart comparing an MBSE approach to traditional SE approaches to evaluate the economic benefits of MBSE. The results indicate that high-stakes industries like transportation, aerospace, defense, and energy are predicted to gain the most from implementing MBSE. After conducting literature reviews, a Likert-style survey, and structured interviews with experts in the fields of MBSE and cost estimation, Balram finds that MBSE "is perceived by those within industry to be beneficial for project management, and … for cost estimations.[16]" While understanding the economic benefits of MBSE may provide valuable information, it may lack a holistic perspective of understanding the larger value of MBSE. Our research will analyze the metrics used to support claims made about MBSE, which will show how authors are measuring the value of MSBE in more ways than just cost.

Studies of the value of MBSE

Three previous studies answered similar research questions to our study.[11,19,22] A recent study by Henderson and Salado[22] searched 20 journals and conferences for any sources that mentioned a benefit of MBSE to understand the value of the approach. With 360 papers containing 1233 benefits, Henderson and Salado found that the vast majority of the benefits of MBSE were perceived, rather than observed or measured. Our study takes into account not only the positive but also the negative aspects of MBSE, whereas Henderson and Salado focus only on the positive aspects (benefits) resulting from the use of MBSE. Both Henderson & Salado's study and our study examine the substantiation of claims made about MBSE. Henderson & Salado grouped the substantiation of benefits into four inductively developed categories: *measured*, *observed*, *perceived*, and *referenced by another source*. In contrast, our study groups attribute and benefit/drawback substantiation into seven inductively developed evidence categories, designed to capture qualitative substantiation (evidence): *Author opinion*, *case study*, *referenced*, *literature review*, *interview*, *survey*, and *community viewpoint*. Regarding quantitative substantiation of MBSE (considered metrics in our study), Henderson & Salado conclude that MBSE benefits are measurable and that metrics can be elicited; they identify only two out of 360 papers that contained benefits with the *measured* claim type. Our study builds on Henderson & Salado's finding that benefits are measurable by assigning metric types (e.g., *cost*, *time*) to claims that provided numerical measurement. Uniquely, our study also utilizes several coders to reduce the inherent subjectivity of utilizing a single coder for thematic analysis.

A study by Carroll and Malins[11] investigated how MBSE was justified, evaluated using a systematic literature review. Carroll and Malins examined a total of 88 case studies, 67 on MBSE without metrics and 21 on MBSE and SE with metrics, to identify whether or not authors

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see MBSE as a justified transition from a DBSE approach. Similar to our study, Caroll and Malins examined the metrics used to justify MBSE. Only 21 of the 88 case studies provided metrics, but only 12 of those case studies were related to MBSE specifically while the other nine were related more broadly to SE. The 67 remaining sources provided attributes without metrics like *completeness*, *consistency*, and *communication of requirements*. Carroll and Malins' findings indicate that "there is a significant advantage to project performance by applying an SE/MBSE approach," but only if it is applied throughout the whole system development life cycle. Carroll & Malins concluded that MBSE improves engineering efficiency and prevents costly rework. Carroll & Malins' research avoids a claim-by-claim analysis of MBSE, choosing rather to summarize individual case studies with metrics to provide insight into MBSE. In addition to case studies, our research will analyze literature reviews, interviews, surveys, etc.–in combination with an analysis of what types of evidence authors are using to make claims about MBSE. With regard to metrics, our study captures measurable claims across all 60 sources, identifying metrics associated with claims both explicitly (directly stated) and implicitly (from context) about *only* MBSE approaches. This is completed through a sentence-by-sentence analysis of MBSE claims in each of the sources.

In a 2018 study, Beaufait[19] created an Excel MBSE analysis tool that provided metrics to assist the Department of Defense program office's decision of whether or not to implement MBSE. The study is unique in its focus on implementing MBSE into a mature program rather than at the beginning of the system development life cycle. To determine if the Excel MBSE tool was successful in recommending whether or not to implement MBSE, Beaufait conducted a side-by-side case study of two inertial navigation systems programs where one uses an MBSE approach and the other uses a traditional DBSE approach. Beaufait's case study provided metrics that support the implementation of MBSE, suggesting that metrics can be used to quantify the value of MBSE for certain applications. While Beaufait conducted a case study, our study provides a thematic coding of a systematic literature review. We also gather perceptions of positive and negative aspects of MBSE, benefits and drawbacks of MBSE, and evidence used to support claims about MBSE in comparison to Beaufait's inertial navigation system case study comparing MBSE and DBSE directly. Our study will further investigate the metrics used to describe MBSE in a larger sample of literature, spanning from pre-implementation to post-implementation of MBSE, rather than the narrower "post Milestone C" focus of Beaufait's study. With 60 academic sources across several different applications (not only defense as with Beaufait's study), our study aggregates metrics to gain an overall perspective of the value of implementing MBSE.

Our study builds upon previous work by identifying the perceived benefits and drawbacks of MBSE in literature and the associated attributes, evidence, and metrics used to measure them.

2 | METHODOLOGY

Our methodology follows six main steps as identified in Figure 1 and described in the following sections.

2.1 | Source identification (Steps 1–3)

Between January 2020 and August 2021, key terms (shown in Table 1) were searched in *Google Scholar*. While there has been some historical debate on Google Scholar's coverage,[23–25] more recent studies have concluded it is a powerful search tool for systematic reviews,[26–29] providing broader coverage than journal searches and prioritizing highly cited works. Fifteen neutral search terms were used to identify sources. Out of the 60 sources selected in this study (listed in Appendix B), 38 were identified with the search term "MBSE."

The search included the title, abstract, keywords, and body text. The sources were not limited by any publication date but were all digitally sourced and publicly accessible. It is important to note that the 60 sources selected were not exhaustive but are composed of highly cited works about MBSE identified through *Google Scholar*. A subset of the results from the *Google Scholar* searches was applicable to this research after the three inclusion criteria[26] described in step three were applied, which are based on practical issues of language, subjects, and publication type. First, only sources published in English were included. Second, a source must have discussed MBSE. Third, non-peer-reviewed grey literature was excluded, including websites, pre-prints, white papers, slide decks, course syllabi, and self-published works. Note that this study did not group conference papers in grey literature, as conferences do have a formal review process. All sources were published between 2002 and 2020.

The specific implementation, tool usage, and methodological preference were not considered as exclusion criteria, as consistent with previous research on the value of MBSE.[22] Although MBSE is an umbrella approach that can refer to any modeling method, our review found that when people describe MBSE and its effects, they were usually describing SysML. Previous studies typically focused on impacts and perceptions after implementation of MBSE; our study also captured impacts and perceptions prior to implementation. Our study identified 17 sources that explicitly describe SysML application, 20 that mention SysML as a part of MBSE, and 23 that describe MBSE in general. This study included both sources using the terms SysML and MBSE under the assumption that when an author described SysML, they were by extension also describing MBSE.

Because this review aims to evaluate the *perception* of MBSE in literature (using the sources themselves as data) rather than aggregate datasets (using data from the sources), quality checks[26] rating the methodology for each source were not completed.

2.2 | Coding and analysis (Steps 4–5)

From 60 sources, a total of 2914 claims describing MBSE were extracted and copied into a database. Claims were included if they described MBSE or an impact of using MBSE. This study utilized a qualitative content analysis methodology—referred to as *coding* hereafter—in order to systematically categorize MBSE perceptions in literature. Coding is considered an efficient, iterative method of analyzing literature and extracting meaningful themes.[30] Two coding processes were

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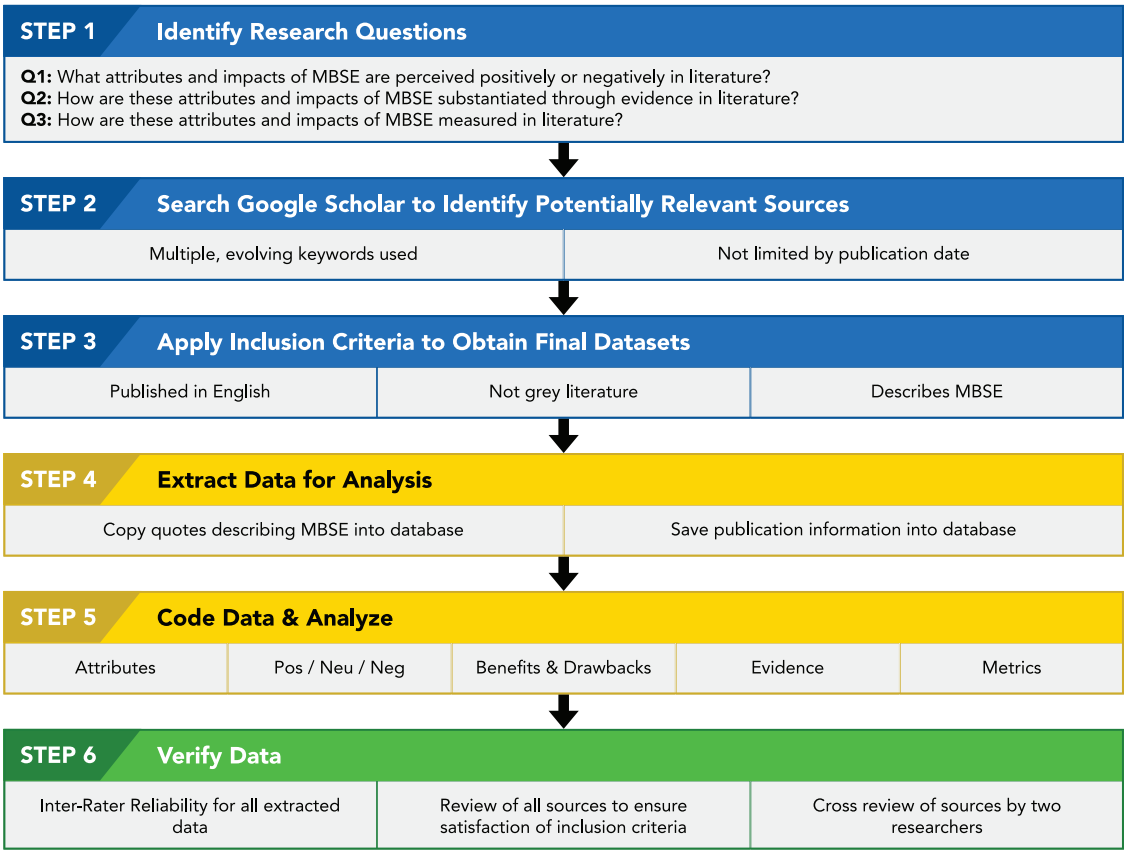


FIGURE 1 Chart details the six-step methodology used in this research

followed depending on if the codes were developed as part of the coding process (inductive) or if the codes were predetermined before the coding process began (deductive).

Inductive coding was required for attributes, evidence, and metrics as no previous coding scheme had been developed. The researchers utilized the constant comparative method for this process, as this is typically seen as the most effective method for analyzing content and developing themes from data.[31] The coding process was open-ended and iterative to reduce bias and discourage the forcing of claims to fit into a predetermined attribute, evidence, or metric list. Multiple researchers were involved in coding the data: preliminary coder(s), final coder(s), and disagreement coder(s). Preliminary coder(s) first assigned an attribute, evidence, and metric to each claim without a predetermined list. After completing the first pass of coding, the preliminary codes were evaluated for redundancies and consolidated to develop a final list of codes, creating definitions for all codes in the process. Next, two final coders independently coded an attribute and an evidence type for each claim using the final list of codes. All disagreements in the attribute, evidence, and metric coding were resolved in discussion between the final coder(s) and disagreement coder(s). The three inductive coding processes are described in Sections 3.2.1 through 3.2.3.

For the deductive codings either a previous coding scheme was adapted (for coding of benefit and drawback) or the codes were self-evident (for coding of positive, negative, or neutral). Three researchers were involved in coding the data deductively: two final coders, and a disagreement coder. As with the inductive coding, two final coders independently coded a positive, negative, or neutral use of the attribute and a benefit or drawback for each claim from the predetermined codes. All disagreements in the attribute coding were resolved in discussion with a disagreement coder. The two deductive coding processes are described in Sections 3.2.4 and 3.2.5.

2.2.1 | Attribute coding

The criteria for the inclusion of an attribute was that it described MBSE or described the impacts of MBSE. Consolidating like and redundant attributes (e.g., consolidating *Time to Develop Model* and *Time to Evaluate Design* into *Time* and combining *Robust* and *Resilience* into *Robustness*) as well as developing more specific versions of certain attributes (e.g., *Understanding* divided into *Approach Understandability* and *Project / System Understandability*) resulted in a list of 68 attributes describing MBSE. This consolidation was an iterative process, requiring the reexamination of all previous coding when any changes were made to the attributes or their definitions, which are shown in Appendix A. Capturing disparate aspects of MBSE can make data analysis more complex due to the large number of attributes used in the coding. It is therefore useful to examine the attributes at a higher level by grouping them into categories based on their similarities. The categories

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are largely subjective, and could be categorized in different ways, but are important for guiding future analyses. The categories are *Communications*, *Implementation*, *Information & Understanding*, *Management & Handling*, *Modernity*, *Performance & Evaluation*, *Resources*, *Structure & Representation*, and *Use*. Their respective attributes are shown in Figure 4.

2.2.2 | Evidence coding

Each claim was coded with a type of evidence supporting the claim based on explicit author statements of evidence type or inferring the type from context. Analyzing the evidence used to make a claim illuminates not only the perceptions of MBSE but the substantiation. Evidence types included: *Case Study*, *Interview*, *Survey*, *Literature Review*, *Author Opinion/Experience*, *Referenced*, and *Community Viewpoint*. The evidence types are defined in Table 3.

Author Opinion/Experience is an evidence type used to support a claim that a source author makes based on their own beliefs or past experiences, whereas a *Community Viewpoint* supports a claim that a source author believes a general community has. For example, if a source author writes, "The MBSE community is pushing for models to replace documents," the evidence would be coded as *Community Viewpoint* because the source's author believes the community has this view about MBSE. Furthermore, *Referenced* is a claim based on a separate source, whereas, *Literature Review* is a claim based on a literature review process that analyzes multiple sources. For example, a claim might state, "21 out of 40 sources reported that MBSE is perceived by those within industry to be beneficial for project management," which would be coded as *Literature Review* because many sources were analyzed to create this claim. A *Case Study* could range from a rigorous experiment with measured results to a single pilot study. A *Case Study* with supporting metrics is the strongest type of evidence a source's author could provide based on this coding scheme. *Referenced* claims could have had varied ranges of credibility depending on whom the source's author cited. This study did not conduct an analysis of the evidence of *Referenced* claims, as this could involve tracing a claim across several papers and was deemed beyond the scope of this study. Similar to *Referenced* claims, *Literature Reviews* also vary as a type of evidence depending on the sources the source's author analyzed. *Interviews*, *Surveys*, and *Community Viewpoints* largely consist of the reported opinions of members of the SE community.

2.2.3 | Metric coding

The criteria for the inclusion of a metric was that it measured a claim about MBSE. Metrics provide a numerical view of MBSE perception, which differentiates metrics from attributes and benefits/drawbacks that provide a qualitative view. Metrics included *Cost*, *Time*, *Risk*, *Number of Errors*, *Number of Defects*, *Number of Model Elements*, *Work Hours*, *Requirements*, *Successes and Failures*, and *Miscellaneous*. *Miscellaneous* included measures like *Effectiveness*, *Number of Iterations*, *Number of Baselines*, *Test Count*, *Probability For System Changes*, *Environmental Impact*, etc. Not all claims had a metric in which case the metric was coded as *no metric*. Claims that involved numerical measurements not related to MBSE were not coded as a metric; metrics had to be used to make a claim about MBSE. For example, a hypothetical "60% of respondents think that MBSE should be implemented in their organizations" would not be coded with a metric.

2.2.4 | Positive, negative, or neutral coding

The codes *positive*, *negative*, or *neutral* were predetermined to assess the attitude of claims. A claim such as "MBSE is more efficient than DBSE," would be labeled positive due to the favorable description of MBSE efficiency. A negative code indicates that MBSE cannot perform the attribute or that it describes MBSE unfavorably. An example of a negative code would be "MBSE tools are not yet mature functionally". A neutral code indicates that a claim was neither positive nor negative about MBSE. A claim would typically be labeled neutral if it made a contradictory or hypothetical claim about MBSE. For example, a statement such as "MBSE could be used as an approach for handling complex systems in the future," would be neutral because it describes MBSE as it could be in the future, not as it is now (i.e., a hypothetical statement).

2.2.5 | Benefit and drawback coding

Each claim was coded with a benefit, drawback, or no benefit/drawback. This coding scheme was adapted from Henderson & Salado[22] to include the corresponding drawbacks. Benefits and drawbacks differ from attributes by only capturing *impacts* of MBSE. For example, a claim such as "MBSE improves the communication between stakeholders," would be coded with the benefit of *Increased Stakeholder Involvement* and an attribute of positive *Communication*. However, for a claim such as "users were able to trace data throughout the model," this would be coded with the positive attribute *Traceability* and *no benefit/drawback* because the claim only references an aspect of MBSE but does not claim that this aspect has a benefit over another approach or method. By utilizing several coders in an open-ended process, our benefit and drawback coding scheme captured impacts of MBSE that were not included in the Henderson & Salado coding scheme. The additional benefits and drawbacks were initially coded as *Other* and then grouped for further analysis in an inductive process.

2.2.6 | Coding summary

Each coding type is summarized in Table 2 with brief definitions clarifying the key differences.

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TABLE 2 Summary of coding types

Coding summary				
Type of coding	**Section**	**Code**	**Intended to capture**	**Codes**
Inductive	3.2.1	Category	MBSE attribute grouping	See Figures 3 and 4
	3.2.1	Attribute	MBSE perceived descriptions	See Figures 4–6; Appendix A
	3.2.2	Evidence	Substantiation of MBSE claims	Author opinion/experience, case study, referenced, literature review, interview, survey community viewpoint
	3.2.3	Metric	Measurement of MBSE claims	See Figure 10
Deductive	3.2.4	Positive / Negative / Neutral	Direction of description/impact of MBSE claims	Positive, negative, neutral
	3.2.5	Benefit / Drawback	MBSE perceived impacts	See Henderson and Salado[22]

2.3 | Verification of the data (Step 6) and analyses

Three verification processes were followed: review of included sources to ensure consistency with inclusion criteria, cross-checking of all 60 sources by two final coders to ensure no relevant claims were excluded, and calculating an inter-rater reliability on all extracted data.

The review of sources involved two reviewers examining the included sources and removing any that violated the inclusion criteria. In addition to performing a second pass of coding for all claims using the finalized coding list, two independent coders also went back through each of the 60 sources to verify that any codable claims missed on the first pass were extracted and included in the database.

An inter-rater reliability statistical calculation, Cohen's kappa, was performed to compare the two independent coders' codings and to verify that the coding was appropriate. The kappa values were calculated prior to the resolution of disagreements. A kappa value of 1.00 represents perfect agreement while a kappa value of 0.00 represents no agreement. A range of 0.61–0.80 is generally recognized as "substantial agreement," while a range of 0.81–1.00 is recognized as "almost perfect agreement".[32] There was an almost perfect agreement for the attributes ($\kappa = 0.845$), as well as for whether the attribute was positive, negative, or neutral ($\kappa = 0.862$). This high inter-rater reliability suggests that the coders understood the claims consistently and that there was no overlap among the attributes and their definitions. An inter-rater reliability statistic was also calculated for the benefits ($\kappa = 0.711$), drawbacks ($\kappa = 0.782$), and evidence ($\kappa = 0.744$) between the two final coders. The Cohen's kappa values reflect high inter-rater reliability for all types of coding, suggesting a strong level of agreement between both independent coders using this study's coding scheme. This yields data with a high level of confidence in the results.

Graphical analysis and comparative analytics were used on the data through methods such as total usage count, percentage distribution, category distribution, and count by source mention to identify higher-level trends and patterns, which allows a more holistic and quantitative understanding of the approach. Chi-square tests for independence were applied when comparing data in the aggregate datasets, resulting in significant relationships between the measured variables. This is elaborated in Section 4.2.4.

3 | RESULTS AND DISCUSSION

3.1 | Research question 1: What attributes and impacts of MBSE are perceived positively or negatively in literature?

3.1.1 | Positive, negative, and neutral perceptions of MBSE: overwhelmingly positive

MBSE is overwhelmingly perceived as positive in literature by the SE community. This perception is evident in the favorable distribution of *positive* claims (79%) in comparison to *negative* (19%) or *neutral* claims (2%). There are over four times more *positive* claims about MBSE than *negative* claims about MBSE, as seen in Figure 2, which shows the total count of each type in the data.

This study provides four hypotheses as to why the claims describing MBSE are overwhelmingly positive, though the hypotheses are by no means exhaustive. First, it is possible that MBSE is beneficial to many applications and the claims largely reflect this idea. MBSE may offer improvements to system development, and authors could be capturing those positive impressions of the approach.

Second, due to MBSE's relatively new emergence in the SE community,[28] researchers may be publishing with more positive attitudes about MBSE due to the pressures to publish on MBSE and to secure research funding. As a result of MBSE's rapid popularity, researchers may believe MBSE will remain a presence in engineering research.

A third potential reason that the claims may be perceived as overwhelmingly positive is that several industries are in the midst of a DE transformation.[33] MBSE is a subset of DE that has become increasingly prevalent at the Department of Defense in part because it is a Systems Engineering-specific methodology.[34,35] An increase in popularity could

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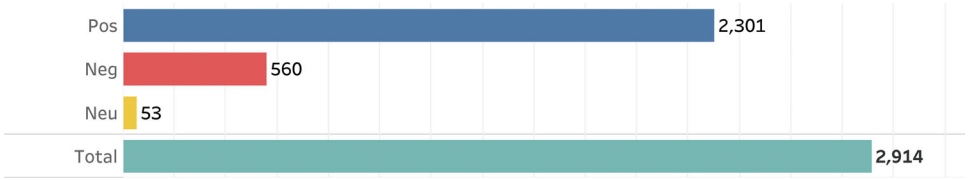


FIGURE 2 Number of positive, negative, and neutral claims

mean an increase in the adoption of MBSE; therefore, there would be more reports of extremely positive or negative results.

Lastly, companies and organizations affiliated with MSBE or MBSE tools may be influencing the positive view of MBSE in the SE community through efforts and campaigns to promote MBSE tools and therefore MBSE. These affiliated companies and organizations include INCOSE, IBM, the OMG, No Magic, Sparx Systems, and Eclipse, among others. INCOSE is currently engaging in an MBSE initiative to "promote, advance, and institutionalize the practice of MBSE.[29]" IBM, which produces the tool Rhapsody, seemingly describes MBSE as a cure-all: "[MBSE is a] highly efficient engineering practice … that improves processes." IBM also encourages the adoption of MBSE, noting that MBSE approaches are "much more reliable than traditional approaches.[36]" The Object Management Group (OMG) is working in conjunction with INCOSE in "helping to establish modeling standards that are essential enablers of MBSE.[37]" No Magic's CATIA MBSE solution is promoted as improving "communications among stakeholders," "improved product quality" through an "unambiguous and precise model of the system," and an "increased ability to manage complexity.[38]" Sparx Systems places emphasis on "complete traceability," "standards based interoperability," "requirements management," and "robust team based systems engineering" in the promotion of its Enterprise Architect MBSE tool.[39] Eclipse promotes its open-source Capella software through "consistency validation," "tradeoffs evaluation," and "reuse of architectures between several projects.[40]"

3.1.2 | Attributes

Figure 3 represents the attributes by category and percentage use. The category bars are sorted into positive and negative uses on a 100-percent scale. The number of attribute uses for each category are labeled on each bar. The *Implementation* category has the most negative perception in the data, as seen by the 58% positive uses (212 claims) and 39% negative uses (153 claims). *Communications* is the most positively-perceived category in comparison to all other categories, with 91% positive uses (273 claims) and 9% negative uses (26 claims). Though *Modernity* is the second-most negative category in the data, there are very few uses throughout (35 total claims).

Figure 4 plots the frequency of each attribute in the data by source mention. This chart counts source mentions as the number of times an attribute is mentioned at least once per an analyzed source, rather than by the number of times an attribute is mentioned throughout all of the data. This is to prevent bias in favor of attributes that are excessively mentioned in a single source. For example, the attribute *Risk & Error Manageability* is mentioned 18 times in a *single* source,[19] which is more than some attributes are mentioned in the whole dataset. If the attributes are sorted by attribute frequency, there are few changes in the top positive and negative attributes when compared with the source mention sorting, though the ordering varies slightly; for example, *Interoperability* and *Compatibility* would switch places with 29 and 24 positive claims, respectively. The only significant difference using attribute frequency rather than source mentions is with the attribute *Cost-effectiveness*; when sorted by attribute frequency, *Cost-effectiveness* is the most mentioned positive attribute in the *Resources* category, whereas, when sorted by source mentions, *Cost-effectiveness* is the fifth most mentioned positive attribute in the *Resources* category. The most common attribute is *Verification & Validation Capability*, with 52 total source mentions (SM) out of a possible 60 sources, and the least common is *Ownership*, with only two SM.

Figure 5 partitions the data into the top ten most frequently occurring positive and negative attributes, respectively, by source mention. The horizontal axis represents the total number of sources that mentioned each attribute at least once. The vertical axis (i.e., the height of the bar) represents the total number of times an attribute is mentioned in a single source, which shows the impact that one source can have on total attribute frequency. The most mentioned positive attributes are *Verification & Validation Capability*, *Consistency*, and *Communication Capability*. The most mentioned negative attributes are *Time*, *Familiarity*, and *Approach Understandability*.

The data show that the most mentioned positive attributes are cited more frequently in individual sources than negative attributes as shown by the taller bar heights (e.g., *Verification & Validation Capability* is positively cited 2.55 times per positive SM on average while it was negatively cited 1.13 times per negative SM on average). This suggests that MBSE literature may be overemphasizing the positive claims about MBSE and understating the negative claims. The following sections discuss some noteworthy findings from the attribute analysis.

Positive perceptions of MBSE capability

The perceptions of MBSE's capability in the SE community are mostly positive, with improved *Verification & Validation Capability* (44 positive SM), *Consistency* (39 positive SM), *Communication Capability* (36 positive SM), *Quality* (20 positive SM), and *Integrability* (35 positive SM) being some of the most frequently appearing attributes positively describing MBSE. The definitions and expected benefits promoted by groups such as INCOSE, IBM, the OMG, Sparx Systems, No Magic,

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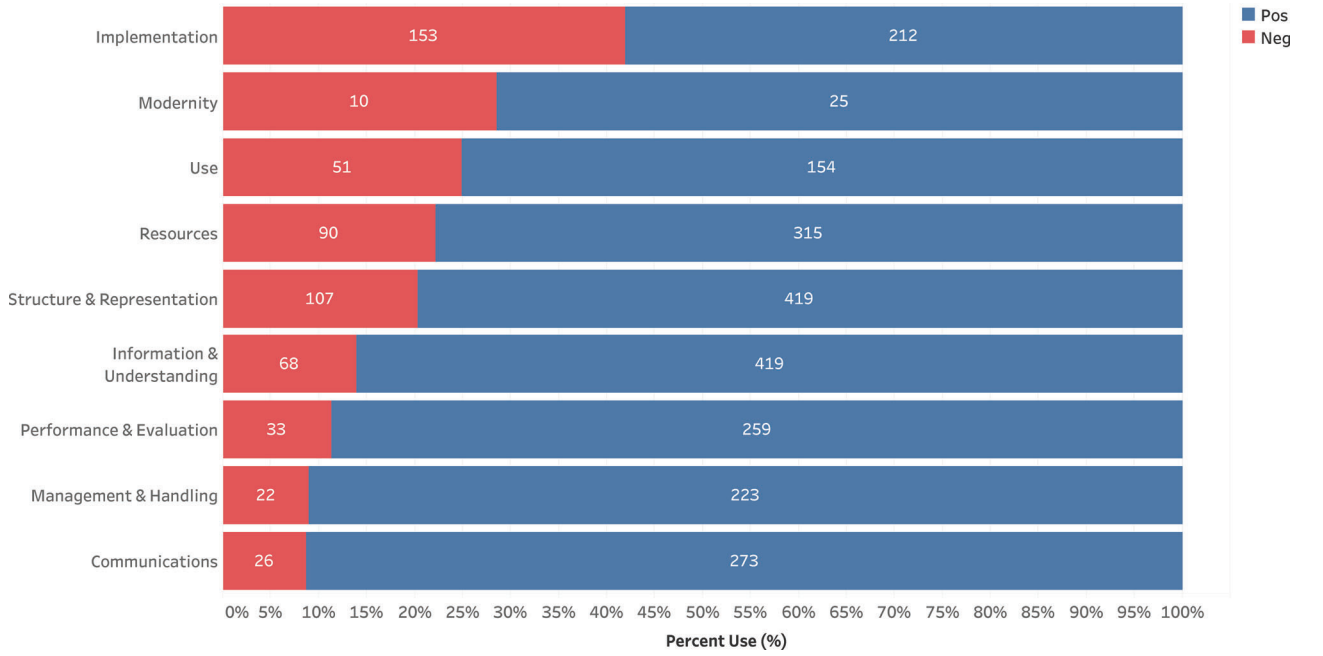


FIGURE 3 Attribute categories by descending negative percentage use

Eclipse, etc.—discussed in 4.1.1—are similar to the most frequent positive attributes. Specifically, INCOSE claims that MBSE "results in quality/productivity improvements & lower risk," due to "rigor and precision," "communications among development team and customer," and "management of complexity" resulting from the approach.[6] IBM, in the promotion of its MBSE-centric Rhapsody tool, claims that models "improve quality, agility, consistency and compliance" in addition to helping "solve the collaboration, cost and calendar problems" faced by systems engineers.[41] Finally, the OMG argues for MBSE's ability of "enhancing productivity and quality, reducing risk, and providing improved communications among the system development team," which is likely in line with INCOSE's own claims due to both being involved in the OMG Systems Engineering Domain Special Interest Group (SE DSIG).[42] It is evident that the promotional efforts may be reflected in literature as several quotes support some of the positive attributes: "MBSE models afford the opportunity to follow more rigorous design and verification processes that require a semi-formal or formal representation of a design[43]" and "MBSE approach made the engineering processes on a complex system development effort more efficient by improving requirements completeness, consistency, and communication.[11]" It is unclear, however, if the literature is parroting statements made by professional organizations, or if the findings are independent.

Mixed-perceptions of some aspects of MBSE

Several attributes cannot be definitively labeled positive or negative due to a significant number of contrasting positive and negative perceptions. In other words, the positive-to-negative ratio falls in the range between 33% and 66%, or the middle third for percentage usage. This range was chosen to identify attributes that had no clear inclination toward describing MBSE as overwhelmingly positive or overwhelmingly negative. Figure 6 depicts the mixed-perception attributes, sorted by increasing positive perception. The attributes include *Mathematical Capability*, *Useability*, *Applicability*, *Scalability*, *Approach Security & Privacy*, *Change Manageability*, *Objectivity*, *System Accessibility*, *Navigability*, *Approach Substantiation*, and *Acceptability*. The 11 mixed-perception attributes could indicate areas of MBSE that are not consistently understood and agreed upon by the SE community. For example, regarding *Applicability*, Beery & Paulo[44] believe there is "potential utility and applicability of the utilization of MBSE to support [Mission Engineering] through an analysis of the early stage design for a naval mine warfare system." This positive claim about MBSE's applicability contrasts with Baduel et al.[45] who claim, "SysML on its own is not the best suited to apply a development method or build meaningful models in systems engineering." In the case of *Applicability*, their disagreements reflect the specificity of the applications discussed.

Another reason for the mixed-perception attributes could be uncertainty or confusion in the MBSE community, likely due to a lack of unified opinions on the aforementioned aspects of MBSE. This may result in a lack of emphasis on these areas of MBSE contribution as demonstrated by the attributes with low amounts of claims such as *Navigability*, *Approach Security and Privacy*, and *System Accessibility*. The low count of claims for *Navigability*, *Approach Security & Privacy*, and *System Accessibility* may show that these aspects of MBSE may be of little importance to the community because they do not have a strong positive or negative community opinion. Additionally, the mixed-perception attributes could be caused by polarized opinions as shown by the high count of positive and negative claims in attributes like *Acceptability* and *Useability*. While there may be many opinions on the *Acceptability* and *Useability* of MBSE and MBSE tools, respectively, the division in positive and negative claims shows that there is not a clear, unified opinion in the SE community.

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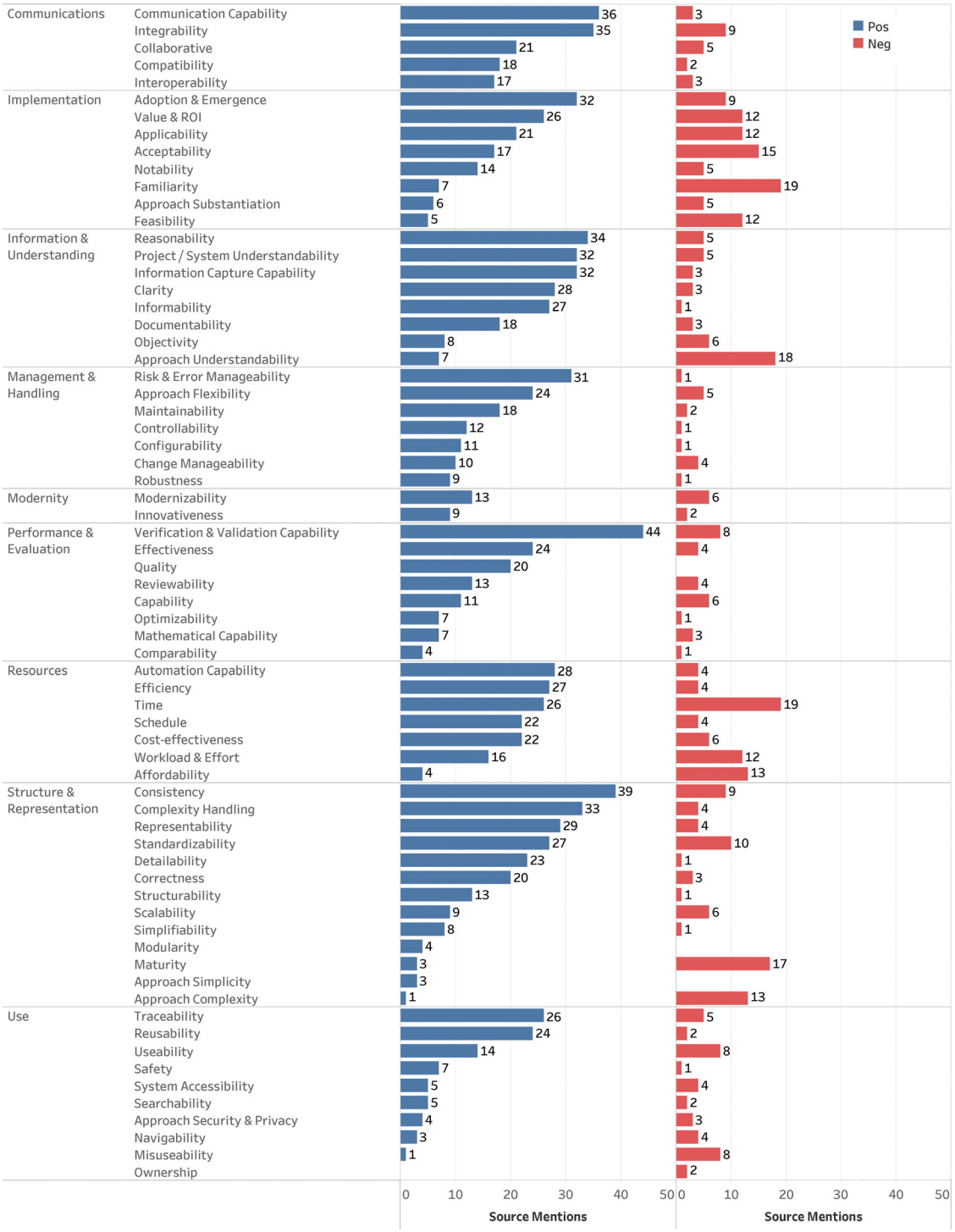


FIGURE 4 Attribute categories with individual attributes sorted by descending positive uses by number of source mentions

Implementing and adopting MBSE: Barriers and enthusiasm

The majority of *Adoption & Emergence* claims were positive (32:9 Pos/Neg SM), which suggests favorability towards the approach among those who have already implemented or are considering implementing MBSE. There was a near-even split in *Acceptability* (17:15 Pos/Neg SM) and a positive-to-negative ratio of 7:18 for *Approach Understandability*. This disparity between a high ratio of positive-to-negative uses for *Adoption & Emergence*, and the lower ratios for *Acceptability* and *Approach Understandability* may produce several conclusions. The first conclusion is that leadership and the SE community may be hesitant to adopt an approach whose benefits are not well understood. The hesitation to adopt MBSE is evident in the even division of the *Acceptability* attribute, which points to confliction in support for MBSE. In addition, the negatively perceived *Approach Understandability* points to a

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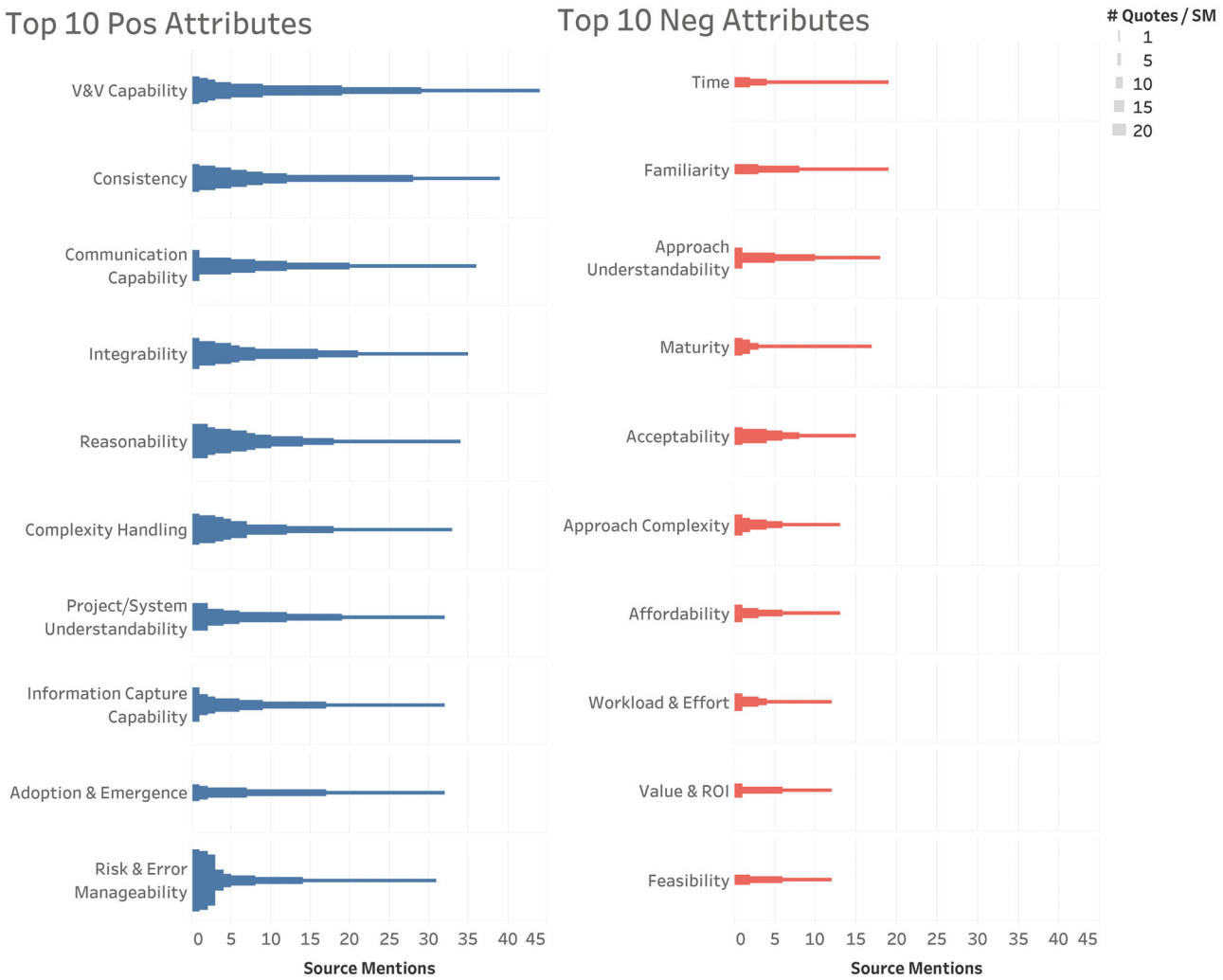


FIGURE 5 Top ten most positively used attributes (left) and top ten most negatively used attributes (right) sorted by source mentions

low understanding of MBSE in the community. By learning more about the approach and its value, the community and leadership may be convinced to implement MBSE. The second factor may have to do with the cost of implementation. This research indicates that the SE community has low positive perceptions of MBSE's *Affordability* (4:13 Pos/Neg SM) and *Feasibility* (5:12 Pos/Neg SM). Furthermore, data suggest that there is a lack of experience and training in MBSE, evident in the low positive perception of *Familiarity* (7:19 Pos/Neg SM) with the approach. Many claims about MBSE are concerned with the amount of training and resources (i.e., time, labor, cost) needed to feasibly implement MBSE. For example, Carroll and Malins[11] state that "an MBSE approach is significantly different from a DBSE approach, forcing changes to engineering processes, tools, and communication methods in order to be successful, and requiring a substantial financial investment for training and tooling."

Few authors dispute the benefits of having implemented MBSE. However, hesitancy to implement MBSE largely centers around a lack of perceived value and applicability to their current projects in comparison to the barriers to implementation. Several sources discuss remaining with a current approach rather than transitioning to MBSE where a perceived return on investment is not guaranteed.[19] "A program that does not continue an MBSE approach throughout the full system development life cycle will likely incur all of the investment cost, but will not reap the benefits or ROI hoped for.[11]" Gregory et al.[1] caution that "the journey from Document-Based Systems Engineering (DBSE) to Model-Based Systems Engineering (MBSE) within an organization can be a long and winding road that requires considerable effort and expense." This hesitancy was reflected in the frequent question of whether the steep costs of implementation and training are worth a perceived marginal return on investment.[11,46–49]

Of the sources mentioning MBSE implementation, nearly 80% positively describe the *Adoption and Emergence* of MBSE. This may suggest that though there are still several issues with cultural change and MBSE implementation costs, the overall perception of the approach is positive with regards to widespread adoption. As cited by several sources, MBSE's ability to handle complexity and to improve decision-making and consistency throughout the lifecycle is a reason for its widespread adoption in aerospace and defense industries.[50,51] In other industries, companies were usually reluctant to adopt MBSE due to the lack of mature toolsets and applications to their particular projects, which is in

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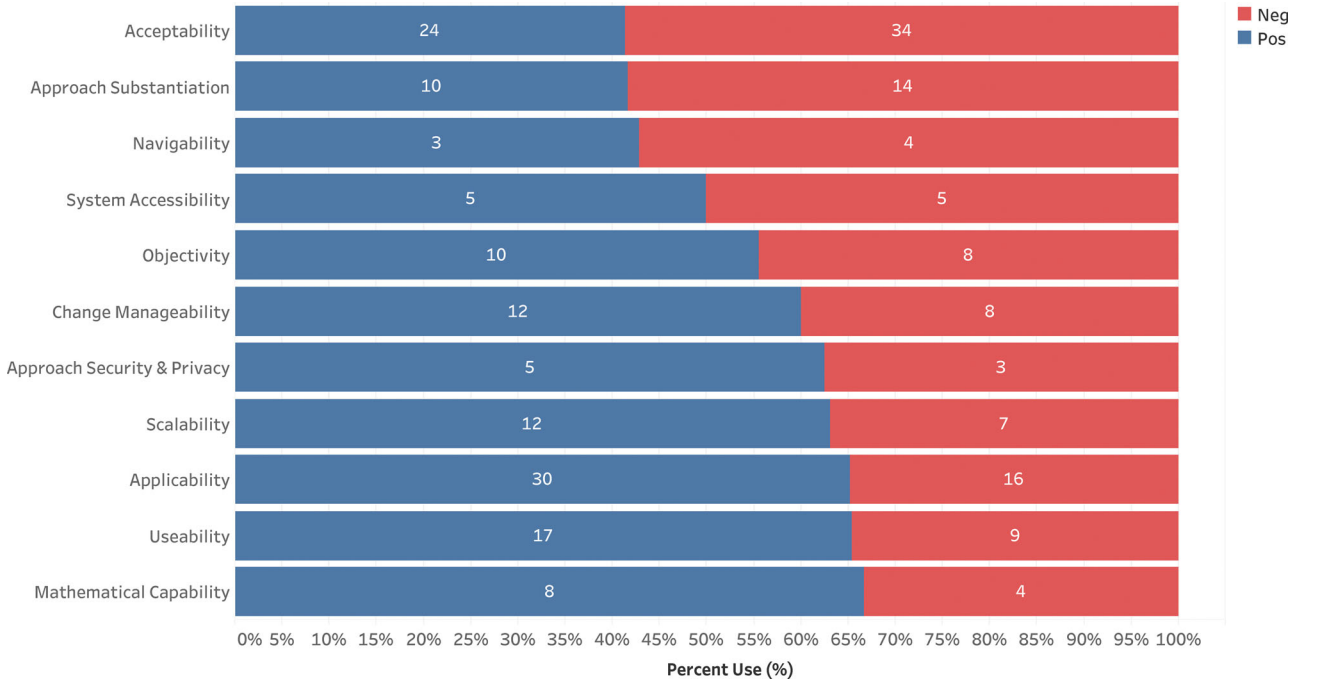


FIGURE 6 Distribution of mixed-perception attributes by descending positive percentage of uses

line with MBSE being an emerging approach.[1,52–55] Despite a few complaints, most of the analyzed sources supported the ability of MBSE to handle increasingly complex projects (33:4 Pos/Neg SM).

Understanding the understandability of MBSE

Published literature praises MBSE for improving understanding of the system for all members of the team (*Project/System Understandability*, 32:5 Pos/Neg SM) through its ability to present information graphically and informatively (*Representability*, 29:4 Pos/Neg SM; *Informability*, 27:1 Pos/Neg SM). However, since the understanding of the approach itself is typically negative (*Approach Understandability*, 7:18 Pos/Neg SM), people may find value in the benefits of the approach without completely understanding the approach as a whole. This is shown in a quote from Owens et al., "The ability of the MBSE tools to create publishable model products without requiring everyone to know MBSE, or its tools, allow for those who wish to use the model, interact with it, and benefit from it without needing to understand the software.[56]" This fits an emergent theme from the analysis, in which MBSE is often only perceived as positive *after* implementation due to high barriers toward adoption. This idea is reflected by Herzog et al., "We also have a paradox in the sense that use of models has yielded positive experiences in all projects where it has been applied … So how could something that is generally perceived as being positive be so difficult to introduce?[48]" Those exploring the potential impact of MBSE should take into account whether the benefit gained for their project application will ultimately outweigh the cost of implementation.

Other potential improvements and disadvantages of MBSE

In analyzing the impacts of MBSE on project *Effectiveness*, this study finds positive perceptions of project improvement from MBSE implementation (24:4 Pos/Neg SM). There is a high positive perception of MBSE for the following attributes: *Informability* (27:1 Pos/Neg SM), *Clarity* (28:3 Pos/Neg SM), and *Project / System Understandability* (32:5 Pos/Neg SM). Furthermore, MBSE is well-suited for flexibility and decision-making, with *Risk & Error Manageability* (31:1 Pos/Neg SM), *Approach Flexibility* (24:5 Pos/Neg SM), and *Reasonability* (34:5 Pos/Neg SM) being described favorably. Finally, the approach has a 33:4 positive-to-negative ratio for *Complexity Handling*. For an approach built around the idea of handling increasingly complex systems, the positive perceptions of *Complexity Handling* positively reflect the commonly cited benefits of MBSE. However, the data point to areas that still need improvement; for instance, the 3:17 positive-to-negative ratio of the *Maturity* attribute. Coupled with high *Misuseability* (1:8 Pos/Neg SM) this may indicate that other barriers to adoption include a lack of refinement and maturity in both the MBSE approach and its tools.

3.1.3 | Benefits and drawbacks of MBSE

Benefits and drawbacks largely align with the top positive and negative attributes of MBSE. However, the ratio of benefits-to-drawbacks is more extreme (6:1) than the ratio of positive-to-negative attributes (4:1). We believe that this may be the result of Henderson and Salado's[22] coding scheme being originally intended to capture only the benefits of MBSE and no drawbacks. This study's attribute coding finds that the many negative aspects of MBSE are perceptions of MBSE implementation, which have no respective codes in the benefit and drawback coding. This is likely due to the idea that benefits are typically emergent after implementation, rather than before.

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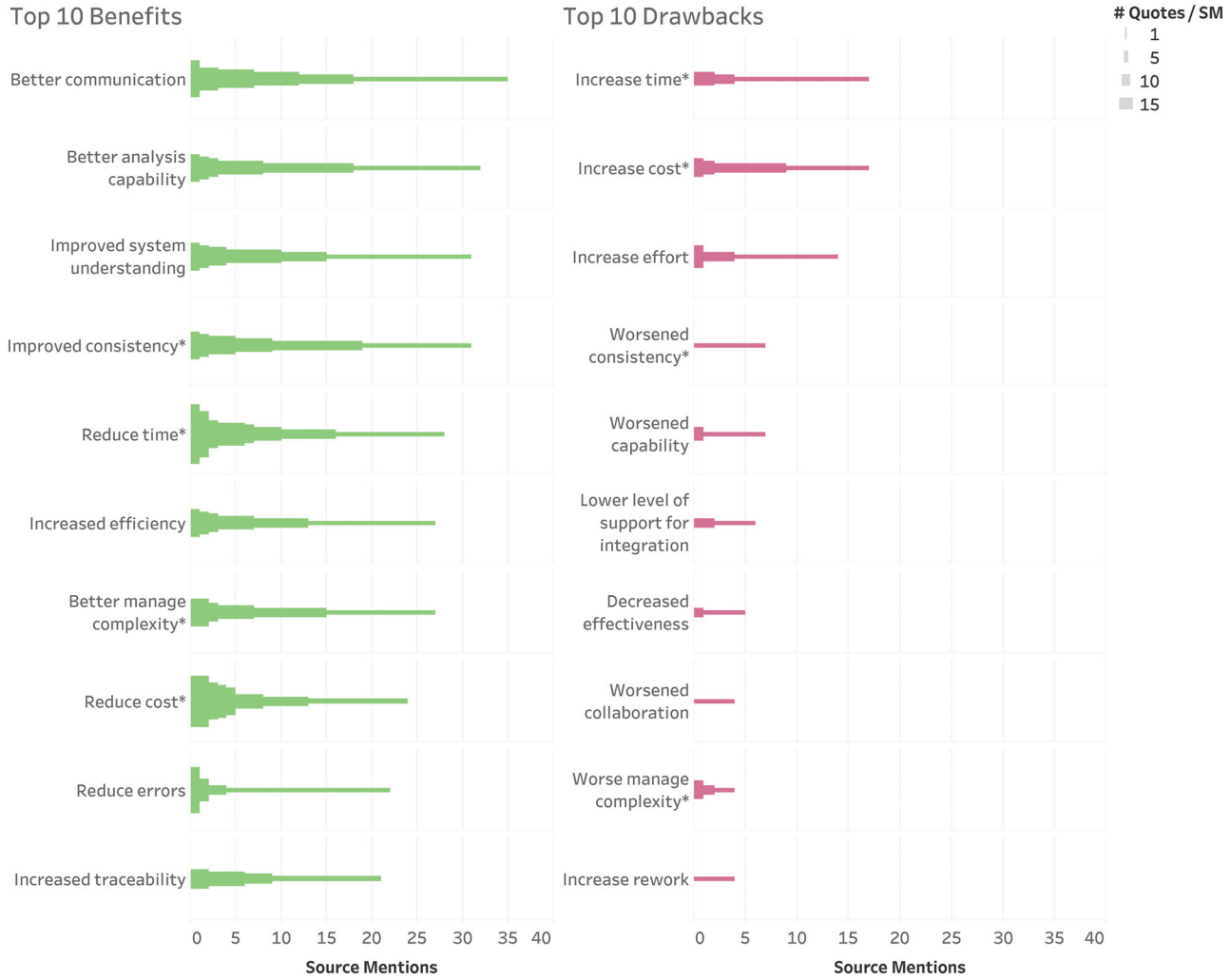


FIGURE 7 Top ten benefits (left) and top ten drawbacks (right) sorted by source mentions

Figure 7 shows the top mentioned benefits and drawbacks and the total number of times a benefit or drawback was mentioned by an individual source, represented by the height of the bar in the vertical direction. Benefits, such as *Reduced time*, are positively cited more frequently in individual sources on average than comparable drawbacks, such as *Increased Time*, which echoes the theme of positive attributes being mentioned more frequently than negative attributes both across sources and within sources. This reinforces the theme that benefits and positive attributes may be more heavily emphasized than drawbacks and negative attributes in literature.

Four of the top 10 benefits are also top drawbacks (i.e., *Time*, *Cost*, *Consistency*, and *Manage Complexity*). The four overlapping top benefits and drawbacks are denoted with an asterisk in Figure 7. The similarities may indicate areas in MBSE perception that may be important to authors of the sources, which may result in their frequent discussion in both positive and negative contexts. While the number of benefits still heavily outweighs the drawbacks, the commonality between the top benefits and drawbacks indicates that there may not be a clear, consistent understanding of the perceived benefits and drawbacks in the SE community. For example, *Reduced Time* is a frequent benefit, yet *Increased Time* is also a frequent drawback. This may be a result of a lack of specificity with the *Time* benefit/drawback (e.g., *time to model the system* is larger, but *overall project schedule* is reduced), or this may suggest that the SE community disagrees on whether aspects of MBSE are definitively benefits or drawbacks, which builds on the earlier discussion of the 11 mixed-perception attributes that were neither definitively positive nor negative. The findings on overlapping benefits and drawbacks may point to a bias toward describing aspects of MBSE through commonly repeated benefits and drawbacks, likely stemming from a generally incomplete understanding of the approach.

In comparison to results from Henderson & Salado's literature review,[22] we found several commonalities between the most mentioned benefits of MBSE. Seven out of 10 top benefits from Henderson and Salado coincided with our top 10 benefits: *Better Communication/Information Sharing*, *Increased Traceability*, *Improved Consistency*, *Better Manage Complexity*, *Reduced Cost*, *Reduced Time*, and *Reduced Errors* (in order of Henderson and Salado's results). The other three benefits found in the top 10 of Henderson & Salado's results that were not identified as top 10 in our data set, included *Increased Capacity for Reuse*, *Improved System Quality*, and *Reduced Risk*. The other three benefits found in the top 10 of our data set that were not identified as top 10 in Henderson & Salado's results included *Increased Analysis*

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Capability, *Increased Efficiency*, and *Improved System Understanding*. The appearance of similar lists of top benefits across separate datasets suggests that our findings may be valid.

"Other" benefits and drawbacks

An *Other* category was created to include benefits and drawbacks not captured in the Henderson & Salado coding scheme, resulting in the addition of 11 benefits and five drawbacks, with a single overlap. The *Other* category excludes benefits and drawbacks with less than three SM. The additional benefits include improved *Useability* (9 SM), *Representability* (8 SM), *Maintainability* (8 SM), *Value & ROI* (7 SM), *Informability* (7 SM), *Interoperability* (6 SM), *Compatibility* (5 SM), *Simplifiability* (4 SM), *Safety* (4 SM), and *Change Manageability* (3 SM). The additional drawbacks include increased *Approach Complexity* (6 SM), and worsened *Useability* (4 SM), *Approach Understandability* (3 SM), *Change Manageability* (3 SM), *Feasibility* (3 SM), and *Maturity* (3 SM). By comparison to the original benefit/drawback list, only *Approach Complexity* would be included in the top drawbacks list (rank 7). The listed benefits capture areas of MBSE related to representation of information and useability, while the drawbacks capture approach complexity and immaturity.

3.1.4 | Research question 1 takeaways

This study found that the top positive attributes and impacts of MBSE centered on effects from post-implementation, such as improved verification and validation, consistency, communication, decision-making, and risk and error management. The top negative attributes and impacts of MBSE centered on pre-implementation perceptions, such as large time investment, low MBSE training experience and understanding, and high cost and effort to implement. MBSE was positively perceived for its capability over DBSE and for its key advantages in collaboration and real-time updates. Despite the finding that the understanding of MBSE as a whole was limited, MBSE was praised for improving the understanding of the project and system for all those involved. Positive attributes and benefits were found to be mentioned more often than corresponding negative attributes and drawbacks in the literature, indicating a possible emphasization of the pros over the cons of the approach. Finally, MBSE was overwhelmingly positively perceived, possibly pointing to tangible benefits in the implementation of MBSE over DBSE and indicating a wave of industry enthusiasm towards MBSE, among other effects.

3.2 | Research question 2: how are these attributes and impacts substantiated through evidence in literature?

3.2.1 | Evidence

The different types of evidence used to code claims in literature are shown in Table 3, along with their *positive*, *neutral*, and *negative* uses.

3.2.2 | Opinions dominate claims about MBSE

Author Opinion/Experience is the most frequently used evidence type to support claims about MBSE. While MBSE may be an emerging approach in the SE community, 47% of the claims made about MBSE are unsubstantiated (i.e., *Author Opinion/Experience*). The second most frequently used evidence type is the *Case Study*, which accounted for 26% of the claims.

3.2.3 | Unsubstantiated benefits of MBSE

After identifying a significant number of positive claims used by INCOSE, IBM, the OMG, along with others like No Magic, Sparx Systems, and Eclipse, this study further investigates which benefits are associated with those claims in addition to the evidence authors use to substantiate their claims. Figure 8 shows the distribution of supporting evidence for MBSE benefits that are frequently mentioned by the organizations and companies affiliated with MBSE and its promotion. Figure 8 also shows the percentage for each type of evidence for promoted benefits in a pie chart. Our analysis shows that the plurality of positive perceptions of MBSE using the promoted benefits is based on *Author Opinion/Experience* at 48%, followed by *Case Study* at 23%, *Referenced* at 19%, and *Other Evidence Types* at 10%. This could indicate that many of the most commonly used benefits of MBSE in the community are unsubstantiated with regard to the already-stated claims about MBSE by major SE companies and organizations. Certain benefits, such as *Better Manage Complexity*, illustrate this point well due to the overwhelming amount of *Author Opinion/Experience* (26) and *Referenced* (16) claims versus *Case Study* (4) claims. However, it is important to mention that while promoted benefits are not significantly more opinion-based than other benefits of MBSE, those listed in Figure 8 are the publicly-promoted aspects of the approach. It would be expected that the promoted aspects of MBSE have more substantiation than other aspects, but it is clear that this is not the case from the data.

As seen in Figure 8, seven out of the 13 INCOSE-, IBM-, and OMG-promoted benefits are in the overall top ten benefits (see Figure 7). Promoted benefits make up a significant portion of all MBSE benefits, whose lack of substantiation has several potential explanations. The first may be that authors are less willing to criticize MBSE in a public setting than in a company/corporate-level MBSE analysis. If this study were to be repeated with private or internal literature and interviews on MBSE, there may be differing levels of substantiation for promoted benefits of MBSE. Another plausible explanation may be that conducting side-by-side comparisons of MBSE with traditional approaches has a steep cost and time commitment, and authors may be operating with a lack of quantifiable data to support their opinions and experiences with MBSE.[11] Authors in public academic literature may also be discussing positive claims about MBSE not because of any substantial evidence, but because of the efforts of MBSE promotion across the industry. INCOSE and others are promoting claims about the positive effects of MBSE which may be resulting in pressure on authors (and possibly unconscious bias in favor of the approach) to

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TABLE 3 Evidence definition, count, and positive, negative, and neutral distribution

Evidence Distribution

Type	Definition	Total	Pos / neu / neg distribution
Author Opinion/Experience	Claims and results that are unsubstantiated or stem from author experience and beliefs	1370	
Case Study	Claims and results from a case or pilot study	750	
Referenced	Claims and results referenced from other sources	519	
Literature Review	Claims and results derived from a literature review of sources	126	
Interview	Claims and results from studies that utilized interviews	81	
Survey	Claims and results from studies that utilized surveys	32	
Community Viewpoint	Claims and results based on reported widely-held community beliefs	17	

repeat promoted benefits and only report definitive drawbacks, possibly to avoid contradicting the current wave of positivity toward the MBSE approach.

3.2.4 | Hesitancy to report drawbacks unless substantiated

For both benefits and drawbacks, Figure 9 compares the evidence used to support the benefits and drawbacks in a percentage distribution chart, as well as the evidence used to support positive and negative attributes. Although MBSE benefits outnumber drawbacks significantly, it is notable to point out that benefits are supported about twice as much by *Author Opinion/Experience* than by *Case Study*, while drawbacks are supported equally by *Author Opinion/Experience* and *Case Studies*. The high count of benefits may be a byproduct of authors being more inclined to speak positively about MBSE and its perceived benefits and hesitant to mention drawbacks unless they are supported by case study data. To verify the benefit/drawback and evidence correlations, a chi-square test for independence was performed on the datasets to determine a relationship between evidence type and benefit/drawback with a p value level of significance of .05. The relation between evidence type and benefit/drawback is significant, $X_4^2 = 25.8$; $p < .001$. Thus, the evidence category seems to be dependent on whether a claim is a benefit or a drawback. It should be noted that four evidence categories with low counts (i.e., *Survey*, *Interviews*, *Multiple*, and *Community Viewpoint*) are combined into a single category called *Other* for the chi-square calculation and future analyses.

Though Figure 9 presents an interesting finding of evidence distribution for benefits and drawbacks, it is important to mention that attributes, when partitioned by positive and negative usage, show much less of a difference between author opinion and case study evidence types. The chi-square test for independence for attributes shows a significant relationship between positive and negative attribute types and evidence type, $X_5^2 = 37.6$; $P < .001$. This decreased difference between benefit/drawback evidence and attribute evidence may be a result of attributes also capturing implementation perceptions, which tended to be more negative. Analyzed literature also tended to be more positive after implementation, possibly explaining the increased difference in the benefit/drawback case study percentages. With this in mind, it seems that positive claims about MBSE may outnumber negative claims significantly due to the frequent citing of opinions

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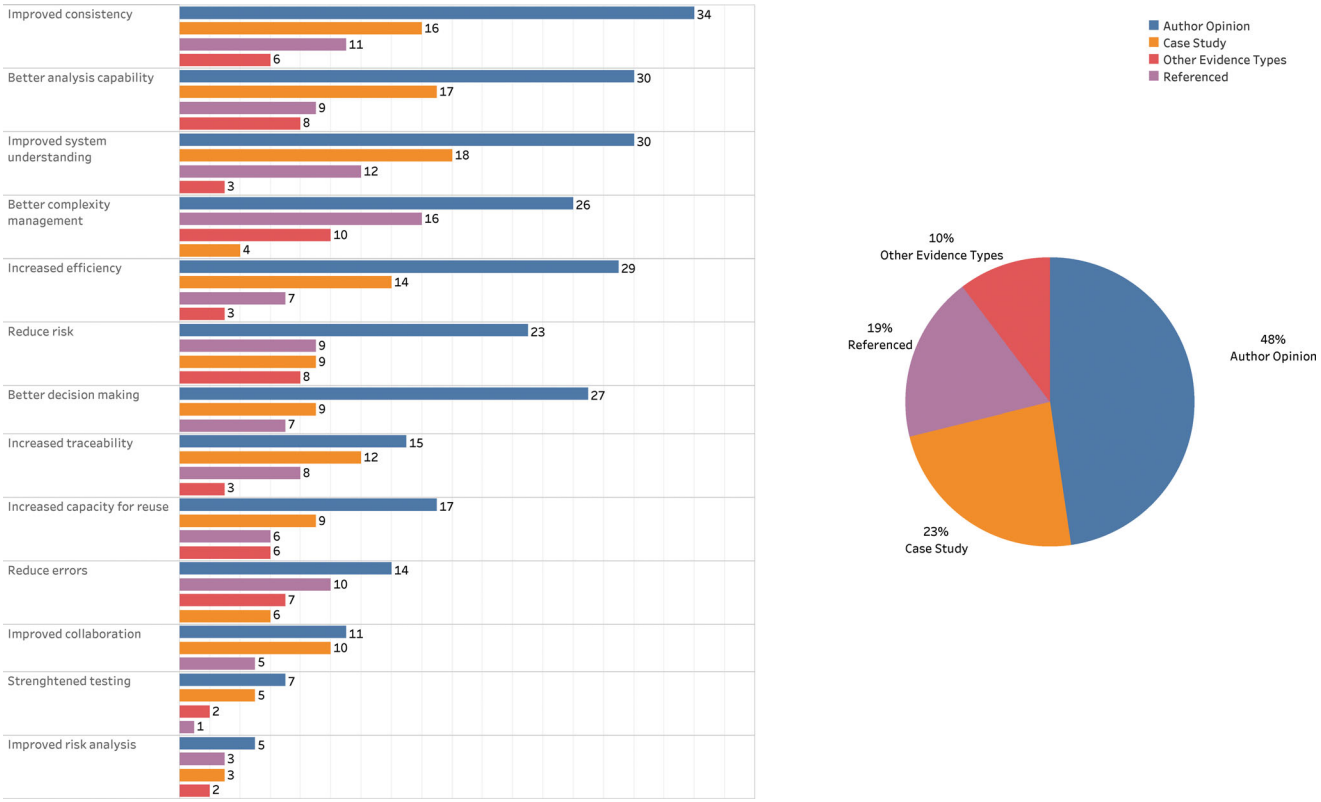


FIGURE 8 INCOSE-, IBM-, and OMG-promoted benefits evidence distribution (left). Overall evidence distribution for INCOSE-, IBM-, and OMG-promoted benefits by percentage

and experiences, rather than relying on case study data, which may be more scarce.

3.2.5 | Research question 2 takeaways

In analyzing substantiation of MBSE claims, it was found that opinions dominated a large part of claims, followed distantly by case studies and other types. The top benefits of MBSE were found to be supported primarily through opinionated claims, including commonly promoted MBSE benefits like improved consistency, better decision making, and better complexity management. It was found that authors tended to emphasize benefits and deemphasize drawbacks of MBSE, with benefits outnumbering drawbacks by a 6:1 ratio. Drawbacks tended to be mentioned more often as part of case study substantiation than benefits, pointing to the possibility of authors being hesitant to mention drawbacks of MBSE unless they were supported by justifiable data.

3.3 | Research question 3: how are these attributes and impacts of MBSE measured in literature?

3.3.1 | Metrics

Figure 10 presents the metrics used to measure claims about MBSE (e.g., *Time*, *Cost*, *Rework*) and whether they are positive, negative, or neutral about MBSE. Metrics that have four mentions or less are coded as *Miscellaneous*, which includes: *Traceability*, *Number of Misunderstandings*, *Number of Vendors*, *Number of Clicks*, *etc.* Only 421 claims (14%) provide or suggest metrics to support their claims about MBSE. By far, most metrics are qualitative (93%) rather than quantitative (7%). The metric findings indicate there is a significant lack of quantitative measurements supporting or rejecting the implementation of MBSE. However, of the claims that do include metrics, the majority (by a roughly 3:1 ratio) are positive towards MBSE.

3.3.2 | Broad cost-benefit of MBSE

The metric of *Cost* is the most used metric to support author claims (145 uses) and measures the cost of using or implementing MBSE; the *Cost* metric often corresponds to attributes such as *Cost-Effectiveness*, *Value & ROI*, and *Affordability*. *Affordability* is one of the top negative attributes of MBSE. However, in Figure 10, the *Cost* metric is used twice as often to support the positive use of MBSE compared to the negative. As stated above, MBSE can have a high cost of implementation, which may not always lead to a positive return on investment. For the *Value & ROI* attribute, there were 57 positive uses describing MBSE and 20 negative uses. This attribute alone is not conclusive in assessing the overall monetary value of MBSE, but it does show that the SE community recognizes a general, unspecified monetary benefit to the approach as a whole.

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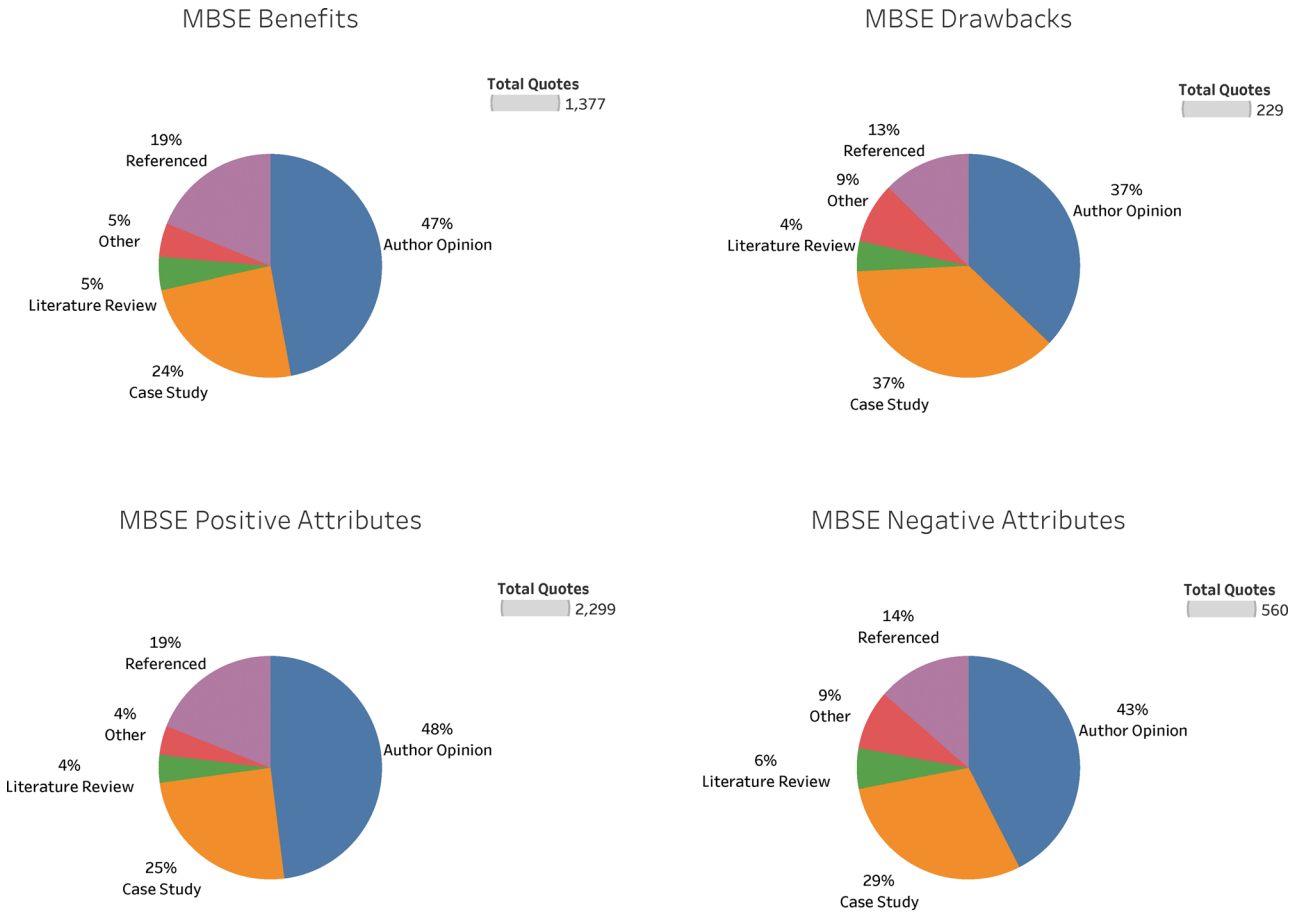


FIGURE 9 Model-Based Systems Engineering perceived benefits (top left), drawbacks (top right), positive attributes (bottom left), and negative attributes (bottom right) sorted by evidence type

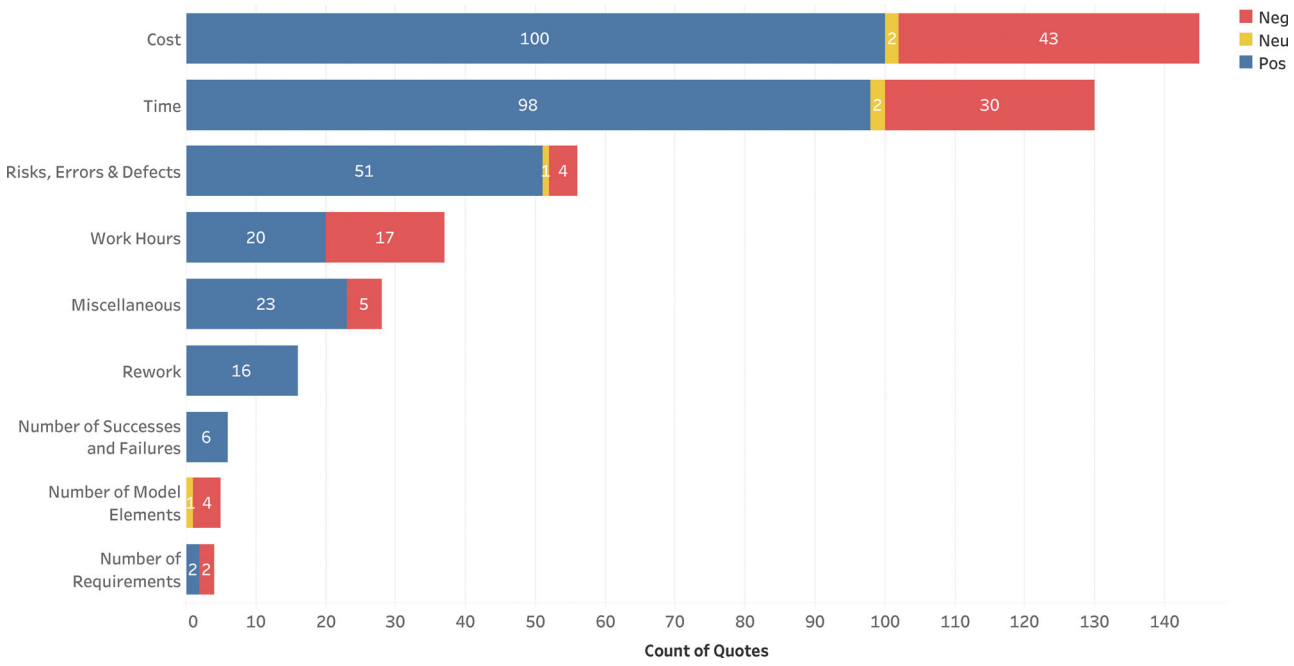


FIGURE 10 Metrics by positive, negative, and neutral counts

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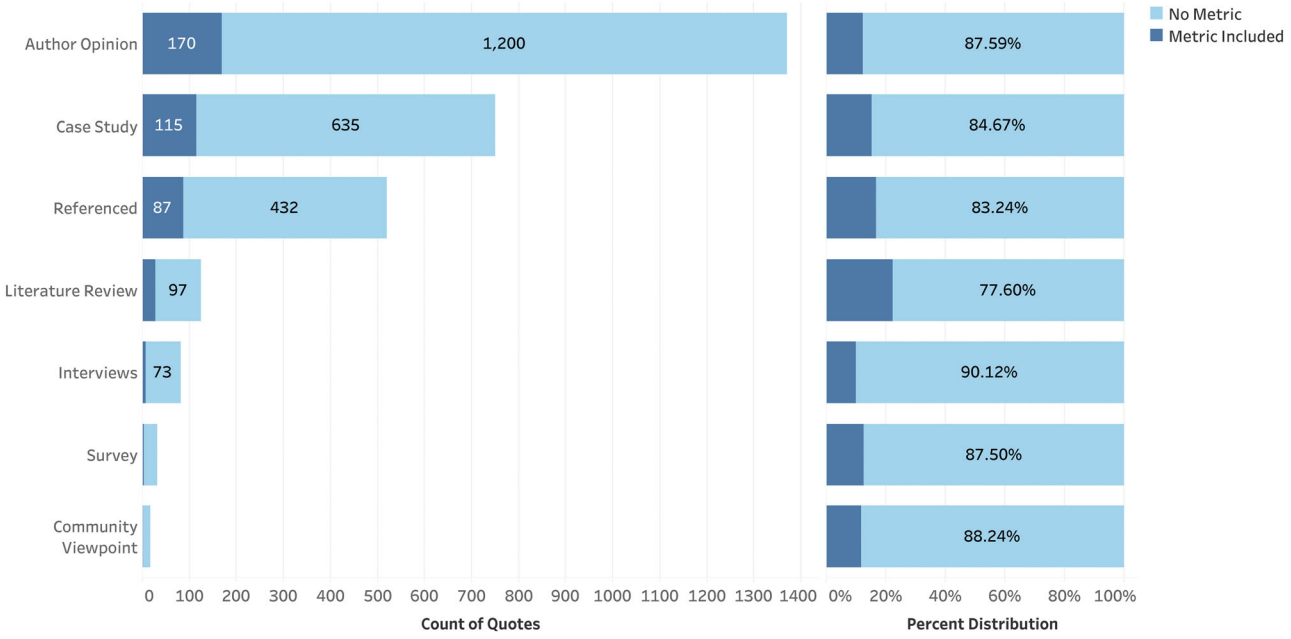


FIGURE 11 Evidence type graphed by total count of claims and further sorted by whether or not a metric was provided for each claim (left). Percentage distribution of claims that did or did not include a metric for each evidence type (right)

3.3.3 | Time advantage for long-term projects

Time is the second most used metric to support claims (130 uses). Analyzing in tandem with the *Number of Work Hours* (37 uses), an interesting pattern arises. The negative mentions of *Time* and *Number of Work Hours* typically describe the length of time needed for the creation of the model, while the positive mentions of *Time* and *Number of Work Hours* largely discuss the reduction in time during the later stages of a project (e.g., verification and analysis phases), which is *after* model development is complete. This may yield the implication, which several authors highlighted, that due to the significant time needed for building a model of the system, implementation of MBSE on short-term projects may not yield much, if any, return on investment.[19] A common recommendation is instead to focus on MBSE implementation in long-term, high-complexity projects due to the benefits down the line. MBSE is often praised for reducing the time for analysis, simulation, verification, and making changes after the model is created since any recognized architectural and system issues could be corrected quickly and implemented through the entire structure simultaneously. One possible reason for this is a lack of training and experience with MBSE creating a barrier to its implementation (*Familiarity*, 7:19 Pos/Neg SM). As a result, projects may need to devote a significant portion of time to training employees on how to properly use MBSE–time that projects may not always have. However, once barriers to implementation are overcome, authors typically find time value from the model post-development activities.

3.3.4 | Strong risk & error management capability

There is a high positive ratio for the *Risks, Errors & Defects metric* (51:1:4). MBSE has the advantageous capability of running simulations of an entire system before any of the system is produced and assembled.[16,35,57] As such, many of the errors that would typically be caught during the testing phase can be caught and rectified earlier in the simulation phase, which in turn saves significant resources for the project. This also improves the reliability and confidence of the hardware version of the model, as the digital twin is typically very similar to the final version of the product, thereby reducing risk. This metric for risk and error management seems to be an area of quantifiable improvement over traditional approaches.

3.3.5 | Evidence-based and metric-based claims

By analyzing metrics and evidence simultaneously, it is possible to see the percentage of claims about MBSE supported by quantitative metrics. The strongest combination of support for MBSE is a *Case Study* with metrics; the weakest is *Author Opinion/Experience* without metrics. Each type of evidence and its associated number of claims with metrics are shown in Figure 11. Note that 19 claims from one source were omitted from Figure 11 due to multiple evidence types being used.

The main takeaway from this data is that although case studies are less numerous than author opinions, the difference between the percentage of quantifiable metrics for case studies and author opinions is likely negligible. Furthermore, there seems to be a similar distribution of metrics per evidence type throughout the data, and this distribution heavily favors *no metric* over *included metric*. This could point to the difficulty of truly assessing the numerical value of MBSE without substantial levels of measurable data, which as mentioned earlier, is expensive and time-consuming to gather. It may take several years and maturation of MBSE tools and products before any evidence-based conclusions can be drawn about the quantitative value of the approach.

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3.3.6 | Research question 3 takeaways

Metrics to substantiate MBSE implementation were scarce, with only 14% of claims including a measure of MBSE. Of the claims that included metrics, authors often stressed a broad cost-benefit to MBSE, time advantage on longer term projects, and stronger risk and error management. A key finding was that quantitative metrics were more scarce than qualitative metrics, likely due to the lack of numerical case study data on the impacts of MBSE.

4 | CONCLUSION

In the MBSE attribute analysis performed in this article, it was found that literature on the topic may be overemphasizing the positive aspects of the approach over the negative aspects. This emphasization could be a direct result of the pressures on the SE community to write positively about MBSE, due to its emergence as *the future of systems engineering*.[13,14] Though strong positive perceptions of system management and communication capabilities were found, negative perceptions of implementation were also identified, likely due to high barriers of adoption. Analyzing the benefits shows that the number of source mentions of benefits highly outweighs the drawbacks in a 6:1 ratio. In combination with the 4:1 positive-to-negative attribute ratio, it may be that benefits and positive attributes are more heavily emphasized than drawbacks and negative attributes in literature. Authors may be choosing to base their descriptions of the approach on what they have heard from groups such as INCOSE and the OMG, even if this is not their intent. Very few claims made about MBSE had quantitative substantiation; though, of the claims that did include metrics, the majority were heavily positive toward MBSE. In short, the community seems to recognize that MBSE has benefits over traditional approaches, but due to a lack of numerical data in publicly available literature and the relative immaturity of the approach, it is unclear how strong the justifications for those benefits are.

Future work involves conducting literature reviews on several other SE approaches utilizing this study's MBSE coding scheme in order to develop an MBSE value model for utilization by practitioners in determining the right approach for their projects. A comparison of MBSE to other SE approaches will be made and each methodology's benefits and drawbacks will be identified. The findings of this study present a strong foundational step toward assessing the true value of MBSE in the SE community, but we believe that future work in the community can build on our findings to further the holistic understanding of the value of MBSE.

This study identifies a need for eliciting project metrics to increase the substantiation of claims made about MBSE. Research into MBSE perception and value could benefit from side-by-side case study comparisons of DBSE to MBSE in which each approach is used for a single project and the outcomes measured and compared. Whether through the use of students as practitioner-surrogates or direct comparison in industry, case study comparison could provide numerical support for the positive and negative aspects of MBSE identified in this study and address the shortage of case studies and metrics identified. Our study also found negative perceptions of *Maturity*, which directly implicates tooling and language limitations. A beneficial venture could be to perform an analysis of MBSE on a tool-by-tool basis, determining each tool and language's benefits and drawbacks to identify sources of negative perception. Finally, it may be useful to analyze MBSE perceptions by specific application to determine whether the project type (e.g., space vehicle vs. atmospheric vehicle) has an impact on community preferences or viewpoints on MBSE.

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[Correction added on 26th March 2023, after first online publication: CRUI-CARE funding statement has been added.]

CONFLICT OF INTEREST

The authors declare that there is no conflict of interest that could be perceived as prejudicing the impartiality of the research reported.

DATA AVAILABILITY STATEMENT

Data available upon request from authors.

ORCID

Kelly X. Campo https://orcid.org/0000-0002-8736-9972
Thomas Teper https://orcid.org/0000-0002-5199-6448
Casey E. Eaton https://orcid.org/0000-0002-7994-077X
Garima Bhatia https://orcid.org/0000-0001-8633-8018
Bryan Mesmer https://orcid.org/0000-0002-4458-6926

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AUTHOR BIOGRAPHIES



Kelly Campo is a junior majoring in industrial and systems engineering at the University of Alabama in Huntsville. As an undergraduate researcher in the Mesmer Research Group, Kelly is focusing on the value of Model-Based Systems Engineering. She has also researched the differences in perceptions of life-cycle models in literature. Kelly plans to pursue a master's degree from UAH. Kelly is a member of Tau Beta Pi and the Rocket City INCOSE Student Chapter.



Thomas Teper is a junior undergraduate student in the Mechanical and Aerospace Engineering program at the University of Alabama in Huntsville (UAH). He is working towards a master's degree in Aerospace Engineering and a minor in Computer Science. In 2019, Thomas interned with Jacobs, working with engineers on the Space Launch System. In the summer of 2022, he interned at Raytheon Missiles & Defense supporting a missile modernization effort and a radar testing program. Thomas currently works as an undergraduate researcher at UAH focusing on approaches to system development, including Model-Based Systems Engineering and life-cycle process models. He is a member of AIAA, INCOSE, and Tau Beta Pi Alabama Delta professional engineering societies.



Casey Eaton is a systems engineering Ph.D. student at the University of Alabama in Huntsville (UAH). Casey's research interests focus on how classification and technical measurement can impact the realization of a system. Casey graduated with her B.S.E. in Industrial and Systems Engineering from UAH in 2019. Continuing at UAH, she graduated with her M.S.E. in Systems Engineering in 2020 with the thesis titled "Failure Classification Schemes: Disciplinary Differences and Trends".



Anna Shipman is a space systems engineer at NASA Marshall Space Flight Center (MSFC), where she is linking her passion for astrophysics and space science with systems development and integration. In 2021, Shipman graduated from UAH with her B.S.E. in Industrial and Systems Engineering and converted from working as a Pathways Student Engineering Trainee at Marshall Space Flight Center to a full-time engineer. Since then, Shipman has been able to work as a systems engineer in support of both astrophysics missions and life support systems.



Garima Bhatia works as a Senior MBSE Engineer at Ford Motor Company and has a background in Aerospace Engineering and Systems Engineering. She received a BS in Aerospace Engineering in India, a MS in Aerospace Engineering from Iowa State University, and a PhD in Systems Engineering from the University of Alabama in Huntsville. Garima has worked on multiple NASA projects in the past, with a focus on MBSE. She leads the modeling effort for NASA's SPORT CubeSat project. Garima is an active member of INCOSE and currently serves as the Vice President/President Elect for the INCOSE Michigan Chapter. She is a past recipient of the Amelia Earhart fellowship.



Dr. Bryan Mesmer is an Associate Professor in the Department of Industrial and Systems Engineering and Engineering Management at the University of Alabama in Huntsville. He completed his Ph.D. in August 2012 at the State University of New York at Buffalo in Mechanical Engineering. His research reimagines systems engineering using approaches that span traditional areas of decision theory and modeling and non-traditional areas of communication arts and psychology.

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APPENDIX

Appendix A: List of attributes and their definitions

List of attributes		
Grouping category	**Attribute**	**Definition**
Implementation	Adoption and emergence	the degree to which MBSE is being implemented by community / company
	Acceptability	likelihood that community / company would want to adopt or implement MBSE
	Notability	distinguishability of MBSE by the community
	Applicability	appropriateness of MBSE for particular community, purpose, or situation
	Familiarity	the degree to which community is experienced with using / can recognize MBSE or MBSE elements
	Feasibility	the ability of MBSE to be practically implemented by the community/company based on metrics of affordability and other factors
	Approach substantiation	evidence supporting implementation and/or benefits of MBSE
	Value & ROI	the worth of an MBSE to a particular community/company in metrics of benefits and return on investment
Structure and representation	Correctness	the degree to which MBSE represents elements/values correctly
	Detailability	the degree to which MBSE represents elements/values precisely, rigorously, and in detail
	Consistency	the degree of consistency and dependability present in MBSE methodology and elements
	Standardizability	the process of making MBSE implementation conform to a community standard
	Maturity	the development status of MBSE technology and methods, including the time in implementation and reduction of system problems and bugs, as well as the development of tools
	Representability	the degree to which MBSE can depict, describe, illustrate system elements
	Structurability	the construction or arrangement of MBSE according to a plan, pattern, or defined organization
	Modularity	the degree to which MBSE or MBSE elements can be divided into subcategories
	Scalability	the ability of MBSE and/or MBSE system to be expanded to other projects, situations, etc.
	Approach complexity	MBSE is difficult to describe, trace, implement, produce products from, etc.
	Approach simplicity	MBSE is easy to describe, trace, implement, produce products from, etc.
	Complexity Handling	the ability of MBSE to mitigate complexity in model design and other project elements
	Simplifiability	the ability of MBSE to represent the system in a simpler way
Management and handling	Controllability	the ability of MBSE to allow for an external input (by a user) to move the internal state of a system from any initial state to any other final state in a finite time interval (i.e., control)
	Configurability	the ability of MBSE to allow for customization of the system to suit project requirements and needs
	Change manageability	the ability of MBSE to manage configurations, their contents, their lifecycles - in particular, identifying and controlling changes to configurations; the ability of MBSE to track and provide control over changes to source code (i.e., version control)
	Approach flexibility	the ability of MBSE to allow for / adapt to project developments over time
	Robustness	the ability of MBSE to respond to errors, inconsistencies, and mistakes in project and system implementation to continue operation at a constant level or to return to pre-change levels

(Continues)

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List of attributes		
Grouping category	**Attribute**	**Definition**
	Maintainability	ability of MBSE to allow for maintenance of the system, including isolating defects or their cause, correct defects or their cause, meet new requirements, make future maintenance easier, or cope with a changing environment, preventing obsoletion
	Risk and error manageability	the ability of MBSE to successfully manage and mitigate risks, and therefore reduce errors
Resources	Efficiency	the degree to which MBSE minimizes costs (labor, monetary, time, etc.) to achieve a desired result
	Affordability	cost of MBSE is the only thing preventing/allowing for implementation by community/company throughout system lifecycle
	Cost-effectiveness	ability of MBSE to create a change in monetary resources
	Schedule	the ability of MBSE to facilitate rearrangement in the current schedule to make the schedule more time efficient
	Time	amount of time it takes for MBSE to perform various processes
	Automation capability	ability of MBSE to reduce human input and workload (i.e., perform tasks autonomously)
	Workload and effort	the amount of effort required from team members using MBSE to achieve expected result
Information and understanding	Clarity	ability of MBSE to make system/ project elements visible to all parties of interest in a way that is coherent and intelligible
	Reasonability	the ability of MBSE to aid in decision making and design based on criteria
	Information capture capability	the ability of MBSE to allow for the process of collecting paper documents, ideas, elements, changes, etc. and transforming them into accurate, retrievable, digital information, and delivering the information into the system model or database for future use
	Informability	ability of MBSE to present useful information about the project/system in a way that is educational, enlightening, etc.
	Documentability	ability of MBSE to allow for project documentation and document generation
	Objectivity	ability of MBSE to develop, present, interpret information without bias
	Approach understandability	ability of people to understand / learn MBSE; understanding of MBSE and its benefits
	Project / System understandability	ability of MBSE to facilitate project and system understanding among team members and stakeholders
Communications	Collaborative	ability of MBSE to promote teamwork
	Communication capability	ability of MBSE to transmit data and information between team members and/or stakeholders
	Compatibility	the ability of two or more systems or components in MBSE implementation to perform their required functions while sharing the same environment
	Interoperability	the ability of MBSE to facilitate the exchange and subsequently use information between two or more systems (or components)
	Integrability	the ability of MBSE to combine two or more components to form an integrated system, which behaves as the system as a whole is expected to behave
Modernity	Modernizability	the conversion and rewriting of legacy approaches, software, and established processes to modern MBSE methods, including programming languages, architecture, software libraries, methods, etc.
	Innovativeness	ability of MBSE to promote new methods, ideas, or products
Performance and evaluation	Capability	the ability of MBSE to execute a specified course of action

(Continues)

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List of attributes		
Grouping category	**Attribute**	**Definition**
	Optimizability	the ability of MBSE to facilitate performance/productivity at the highest efficiency
	Effectiveness	the degree to which MBSE is successful at producing desired results
	Comparability	the ability of MBSE to analyze the differences between system elements/evolution over time
	Mathematical capability	the ability of MBSE to allow for mathematical assessment
	Reviewability	whether the internal state variables of the MBSE system can be externally measured, such as approval, system parts, monitoring, etc.
	Quality	the ability of MBSE to improve the caliber of system elements and/or the final product
	Verification and validation capability	the degree to which MBSE allows for verification, certification, testing, validation, and analysis of system, data, etc.
Use	Navigability	the ease by which MBSE system models can be traversed
	Misuseability	the ease by which MBSE can be incorrectly used
	Searchability	the ease by which MBSE allows system to be searched
	Useability	the degree to which MBSE allows specific user in a specific context to use a product/design to achieve a defined goal effectively, efficiently and satisfactorily
	Traceability	capability of MBSE to trace course of development from origin to current status; ability to verify the history, location, or application of an item by means of documented record identification
	Reusability	the ability of MBSE to allow for the reutilization of system elements and architecture
	Ownership	the degree to which MBSE relies on model ownership to moderate system access, modification, and integrity of elements
	Approach security and privacy	the ability of MBSE to protect against unauthorized system function or software service alterations through abstraction, encryption, etc.
	Safety	the ability of MBSE to protect against danger, risk, or injury
	System accessibility	the ability of MBSE to allow information sharing about system between various levels of clearance and/or outsiders

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